DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL

FINAL EXAMINATION
Course Title: Artificial Intelligence
Course Code: CSE-3205
3rd Year 2nd Semester
Admission Session: 2021-22

Time: 3 hours **Marks:** 60

Answer any five Questions from the followings.

1. a) Define an Intelligent Agent. Explain the role of Sensors, Actuators, and the Environment in designing an intelligent agent. [4]

b) A smart irrigation system automatically waters crops based on soil moisture, weather conditions, and temperature. Describe the structure of this intelligent agent and explain how it makes rational decisions. [4]

c) Differentiate among the following agent types with one real-world example of each. [4]

- Simple Reflex Agent
- Goal-Based Agent
- Learning Agent

2. a) A robot is delivering medicines inside a hospital. The rooms are Entrance (E), Reception (R), Pharmacy (P), Laboratory (L), ICU (I), Ward (W), Exit (X). The corridors are described as follows. [5]

- Entrance connects to Reception.
- Reception connects to Pharmacy and Laboratory.
- Pharmacy connects to ICU.
- Laboratory connects to Ward.
- ICU connects to Exit.
- Ward also connects to Exit.

Draw the graph based on the description. Perform Breadth-First Search (BFS) from Entrance (E) to Exit (X) and show Frontier Queue, Visited Nodes and Parent Table after each iteration.

b) Using the same graph, perform Depth-First Search (DFS) from Entrance (E) to Exit (X). Show the exact order of node expansion. [5]

c) Compare BFS and DFS in terms of Completeness and Optimality. Explain why they may return different solution paths. [2]

3. a) A delivery company wants to transport goods from Warehouse (W) to Customer (C). Locations are : W, A, B, D, E, C. Road costs are [5]

- $W \rightarrow A = 5$
- $W \rightarrow B = 2$
- $A \rightarrow D = 3$
- $B \rightarrow D = 4$
- $D \rightarrow E = 2$
- $E \rightarrow C = 3$

Draw the weighted graph and Perform Uniform Cost Search (UCS). Show Priority Queue, Expanded Node, Current Path Cost after every iteration.

b) Show the final optimal path. [2]

c) Explain the advantages of Iterative Deepening Search (IDS) over BFS and DFS. Also compare their Time Complexity and Space Complexity [5]

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4. a) A robot must travel from Start (S) to Goal (G). The nodes are S, A, B, C, D, G [6]

The path costs are S → A = 3, S → B = 2, A → C = 4, B → D = 2, C → G = 3, D → G = 5. The heuristic values are

Node	h(n)
S	8
A	6
B	5
C	3
D	4
G	0

Draw the graph. Perform A* Search and show: g(n), h(n), f(n), OPEN List, CLOSED List after every iteration.

b) Write the final optimal path found by A*. Using the same graph, perform Greedy Best-First Search (GBFS). Show Priority Queue, Visited Order, Final Path. Compare the result with A* Search. [6]

5. a) Define a Heuristic Function. Explain the characteristics of an admissible heuristic with an example. [3]

b) Using the graph in Question 4, perform Weighted A* Search with [6]

$$f(n) = g(n) + 2h(n)$$

Compare the search process with ordinary A*.

c) Explain the concept of Local Search Algorithms. Why are Local Search algorithms preferred for solving large optimization problems? [3]

6. a) Explain the differences among: Supervised Learning, Unsupervised Learning, Reinforcement Learning based on Training Data, Learning Objective and Output. Give one application of each. [8]

b) For each application below, identify the most suitable learning paradigm. [4]

- Medical disease diagnosis
- Customer segmentation
- House price prediction
- Robot navigation
- Email spam filtering
- Movie recommendation

Also specify whether the task is Classification, Regression, or Clustering, where applicable.

7. a) A machine learning dataset contains the following problems. [6]

- Missing values
- Duplicate records
- Outliers
- Features with different scales
- Categorical attributes

Explain appropriate preprocessing techniques for handling each problem.

b) Explain the one-hot encoding and label endoing techniques with examples. When should each technique be used [6]

8. a) A hospital wants to develop a machine learning system to predict whether a patient has diabetes based on age, BMI, blood pressure, glucose level, and family history. Describe the complete Machine Learning process for developing this system. Your answer should include: [6]

- Data Collection
- Data Preprocessing
- Feature Selection
- Model Training
- Model Evaluation
- Model Deployment

b) Explain the importance of Feature Selection in Machine Learning.Describe any three feature selection techniques. Also explain how feature selection differs from dimensionality reduction. [6]

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